

Dharmendra Ahirwar

Senior Software Engineer | Platform & Infrastructure

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Summary

Backend and platform engineer with 4 years of experience building production systems in Python. Currently sole owner of the technology platform at a regulated digital-assets quantitative hedge fund: 15+ production services across an 11-host fleet, spanning architecture, distributed systems, real-time data infrastructure, reliability engineering, security and release engineering. Track record of taking systems from prototype to supervised, observable, documented production platforms, and of owning incident response end to end.

Skills

Languages: Python, SQL, Java, C++, C, JavaScript, Shell Scripting

Backend & Async: FastAPI, Django, asyncio, Pydantic, Celery, REST APIs, WebSockets, OpenAPI v3, JWT, OAuth 2.0, Unicorn, Uvicorn, Nginx

Data & Storage: PostgreSQL, Redis (Sentinel, Lua scripting), DuckDB, MySQL, MongoDB, SQLite, Amazon RDS, asyncpg, SQLAlchemy, streaming replication, query and schema optimisation

Orchestration & Streaming: Apache Airflow (CeleryExecutor, active-active HA), Apache Kafka, RabbitMQ, real-time WebSocket ingestion, event-driven pipelines

Infrastructure & DevOps: Linux, Docker, systemd, GitHub Actions, CI/CD, Terraform, Ansible, Kubernetes, Helm, Azure DevOps, Jenkins, AWS Secrets Manager, Microsoft Azure, Tailscale, UFW

Observability & Reliability: Prometheus, Grafana, Alertmanager, Elasticsearch, Logstash, Kibana, SLOs, high availability, disaster recovery, incident response, root-cause analysis

Practices: System Design, HLD/LLD, Distributed Systems, Microservices, Design Patterns, SOLID, Concurrency, Caching, Load Balancing, Scalability, Security Hardening, Test Automation, Technical Documentation, Mentoring

Experience

Software Engineer, Hillroute Capital | Jan 2025 – Present, Delhi, India

Sole owner of the technology platform for a regulated digital-assets quant fund.

- Built the firm's **real-time market-data platform** from scratch in async Python/FastAPI across three exchanges (Binance, OKX, Deribit): WebSocket trade-level streams with REST validation and backfill, dollar-bar aggregation pipelines and **530+ symbols**, cutting data lag by **over 80%** and materially reducing execution slippage.
- Engineered correctness guarantees into ingestion — Lua-scripted atomic gap detection enabling redundant staggered instances, dual-path WebSocket/REST reconciliation at **0.01% tolerance**, and a freshness contract that downstream strategies poll before reading, so no strategy ever executes on stale or incomplete data.
- Designed and shipped a **centralised exchange API gateway** with a Redis-Lua distributed rate limiter and multi-egress, weight-aware failover: **150 endpoints, 89 automated tests** across a five-tier pyramid (unit, integration, load, chaos), eliminating cross-service rate-limit contention and exchange-side bans.
- Upgraded **Apache Airflow 2.9 to 3.1** and re-architected it to **active-active high availability** — **105 DAGs**, 8 routing queues, 7 Celery worker hosts — with deploys that gracefully drain in-flight tasks so a release can never interrupt a live trade; migrated cron-based workflows into Airflow for visibility and auditability.
- Executed a **zero-downtime migration of the entire PostgreSQL estate** to new hardware, and added a **streaming hot standby** sustaining **~40 ms** replay lag, closing the platform's largest disaster-recovery gap.
- Built a **DuckDB read tier** mirroring a **60M-row** dataset so research and backtesting workloads never contend with production trading.
- Took the organisation from **no CI/CD to 12 automated pipelines** on a shared, resource-capped self-hosted runner pool, plus a staging environment with fail-closed production isolation and full promotion-to-production verification.
- Implemented **fleet-wide observability** (Prometheus, Grafana, Alertmanager, custom exporters across 11 hosts), zero-trust private-network architecture, host-hardening baselines, and unified failure alerting with phone escalation for risk-critical jobs.
- Authored the organisation's engineering conventions and **~31,000 lines** of architecture, runbook and operational documentation; mentor to a junior engineer.

Senior Software Engineer, Helloverify India Private Limited | Jul 2022 – Aug 2024

- Rebuilt core services in Python/Django, replacing a legacy system and **increasing throughput efficiency by 200%**.
- Architected a migration to microservices to resolve scaling limits, **reducing downtime by 40%**; optimised database schemas and indexing for a further **40% performance gain**.
- Tuned Unicorn, Nginx, Celery and Redis configuration, **reducing latency by 70%** and improving resource utilisation.
- Automated CI/CD with Azure DevOps (**50% faster deployments**) and containerised services with Docker, **cutting cloud costs by 30%**.

- Embedded quality and security gates — SonarQube for PEP 8 enforcement (**30% fewer code smells**) and SOOS for SCA, SAST and DAST — and deployed Prometheus/Grafana monitoring for **40% faster issue detection**.

Software Engineer Intern, Pirates India | May 2020 – Aug 2020

- Designed database schemas for **50,000+ daily transactions**, reducing query time by **30%** across **100,000+** records through appropriate constraints and indexing.
- Built a Python machine-learning recommendation engine that improved content relevance and increased user read-time by **25–30%**.

Education

Bachelor of Technology (B.Tech), Mathematics and Computing | Jul 2018 – Apr 2022

Indian Institute of Technology Delhi (IIT Delhi), India

Achievements

- Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship, SX Stream — 2018.
- Qualified JEE Advanced; admitted to IIT Delhi — 2018.
- Dakshana Foundation Fellowship, two years of JEE coaching — 2016.